Saudi Customs Traffic Data Analysis

Data Sources and Analysis Tools

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1. Introduction

This document outlines the data source and the main tools used to analyze customs traffic across Saudi land ports. The analysis focused on identifying differences between ports, seasonal patterns, and peak operating hours. The results were then used to support traffic forecasting and the development of an AI-powered decision-support prototype.

2. Data Source

The dataset was obtained from the Saudi Open Data Platform and was provided by the Zakat, Tax and Customs Authority (ZATCA). It contains historical customs traffic records for Saudi land ports and includes information about the port, date, hour, direction, and number of movements.

- Dataset: Saudi Customs Traffic Data (Land Ports)
- Coverage: 11 Saudi land ports
- Time period: 2024-2025
- Size: 72,272 hourly records
- Main fields: Port, Date, Hour, Direction, and Number of Movements

This dataset was selected because it is based on real operational data and provides enough detail to compare customs traffic by location and time.

3. Data Preparation

The data was cleaned and prepared before the analysis stage. column names were standardized, and the date field was converted into a consistent datetime format. Missing values and inconsistent records were also checked to improve data quality.

Additional time-based features were created, including year, month, quarter, week, day of the month, and day of the week. These features made it possible to study traffic patterns from different time perspectives. Daily and monthly totals were also prepared for the forecasting stage.

4. Tools Used

The analysis and prototype development were completed using the following tools:

- Python was used as the main programming language for data preparation, analysis, and forecasting.
- Pandas and NumPy were used to clean the dataset, create new variables, group records, and calculate traffic summaries.
- Matplotlib was used to create visualizations for port traffic, monthly patterns, hourly activity, and forecast results.
- Prophet was used to build the time-series forecasting model.
- Streamlit was used to present the analysis, forecasting results, and AI assistant in one interactive application.

5. Analysis Performed

Exploratory Data Analysis (EDA) was carried out to understand the main patterns in the dataset. The analysis focused on traffic differences between ports, changes across months, and hourly activity.

Port analysis. The total number of movements was calculated for each port. Al Batha recorded the highest traffic with 757,722 movements, while Al Durrah recorded the lowest with 10,112 movements. This result shows that operational demand varies significantly between locations.

Monthly analysis. Traffic was grouped by month to identify changes over time. October 2025 recorded the highest monthly total with 127,686 movements, showing that customs activity is not consistent throughout the year.

Hourly analysis. Traffic was also grouped by hour. The highest level was recorded around 4:00 PM, with 105,570 movements across the analyzed period. This finding can help identify when additional staff or resources may be required.

6. Forecasting

For forecasting, daily traffic totals were prepared as a time series and divided into training and testing periods. Prophet was then used to predict future customs traffic. The model achieved a MAPE of 9.44%, an MAE of 979.05 movements, and an RMSE of 1,167.98 movements.

These results indicate that the model can provide useful estimates of future traffic and support operational planning and resource allocation.

7. Supporting AI Tools

FAISS, Hugging Face embeddings, LangChain, and Google Gemini were used to build a Retrieval-Augmented Generation (RAG) assistant. The assistant retrieves relevant customs records and generates answers based on the available data. This feature was added to make the analysis easier to access through natural-language questions.

8. Conclusion

Overall, the analysis used Saudi customs traffic data to identify differences between ports, detect monthly and hourly patterns, and build a forecasting model. Python, Pandas, NumPy, Matplotlib, and Prophet were the main tools used for the data analysis, while Streamlit and the RAG tools were used to present the results in an accessible application.